Accuracy and Imbalance

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Responsible Plant Classifier

Before you start

Week 10: a majority baseline may miss every rare positive.

Plan time to read, investigate and explain. Keep predictions separate from observations.

Your mission

> Start with the target: Explain why overall accuracy can hide failure on a less common class. Try the recall before opening notes.

Your learning target

I can explain why overall **accuracy** can hide failure on a less common class.

I can compare a **training-majority baseline** with class-specific results.

I can calculate an equal-class average and report its limits alongside the confusion matrix.

Remember before reading

Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

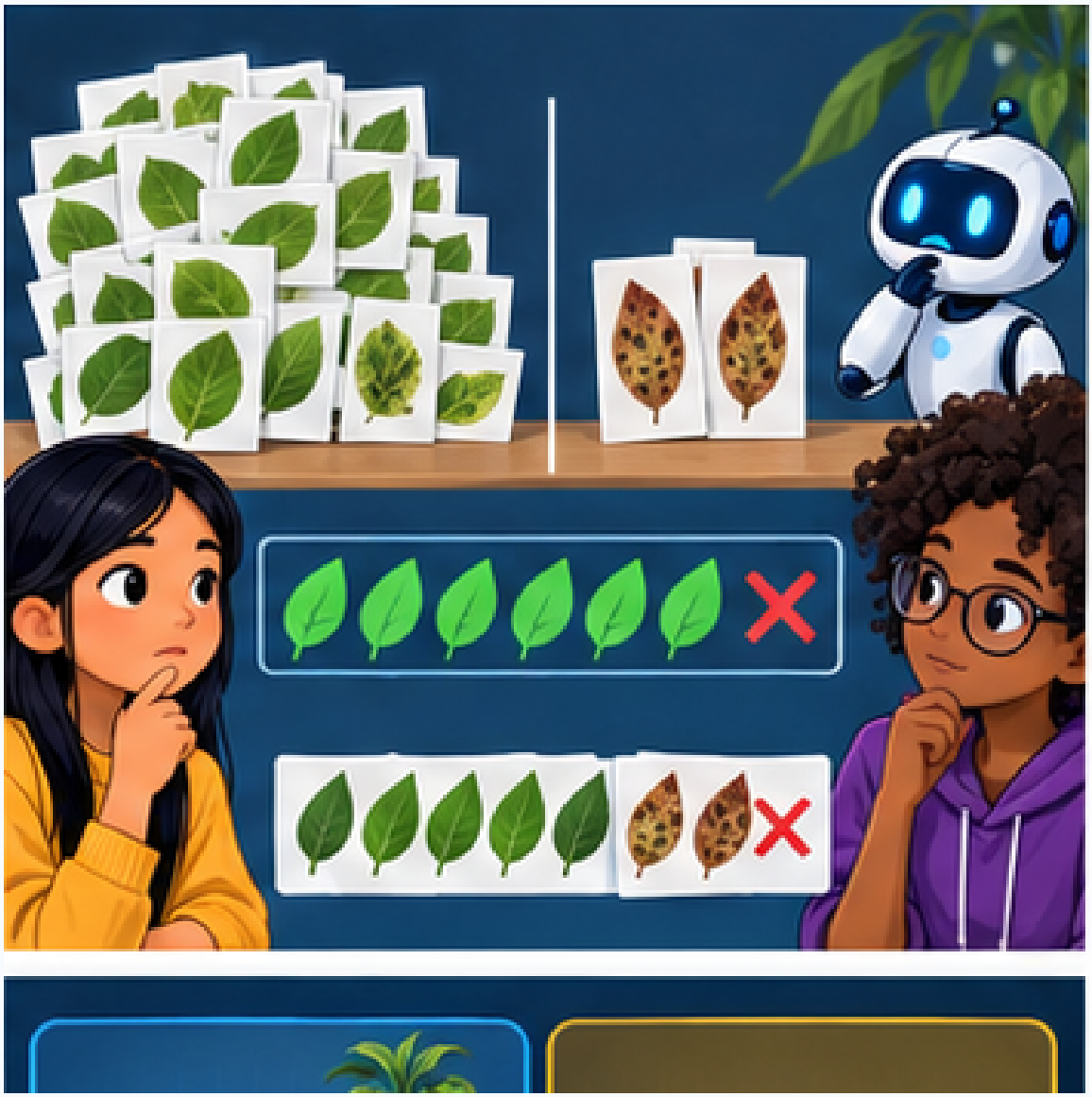
Without opening the old lesson, explain one key idea from Confusion Matrix. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Train a Classifier. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Baselines First. Give an example and a mistake that the idea helps you avoid.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the **evidence** shows was wrong.

Accuracy can be dominated by a common class and hide failure on a rare but important class.



Your target

LEARNING TARGET Model A is always healthy on 96 healthy and 4 spotted leaves. Model B finds 3 spotted but falsely flags 5 healthy. Compare **accuracy** and spotted detection.

Builder name and experiment date:

Readiness for the connected explanation

With reference rows and **prediction** columns ordered clear then blocked, where are false negatives when blocked is positive?

This week's end goal

By the end, I can explain **accuracy** and imbalance using a traced example and a stated limitation.

STUDENT LESSON · 2 OF 18

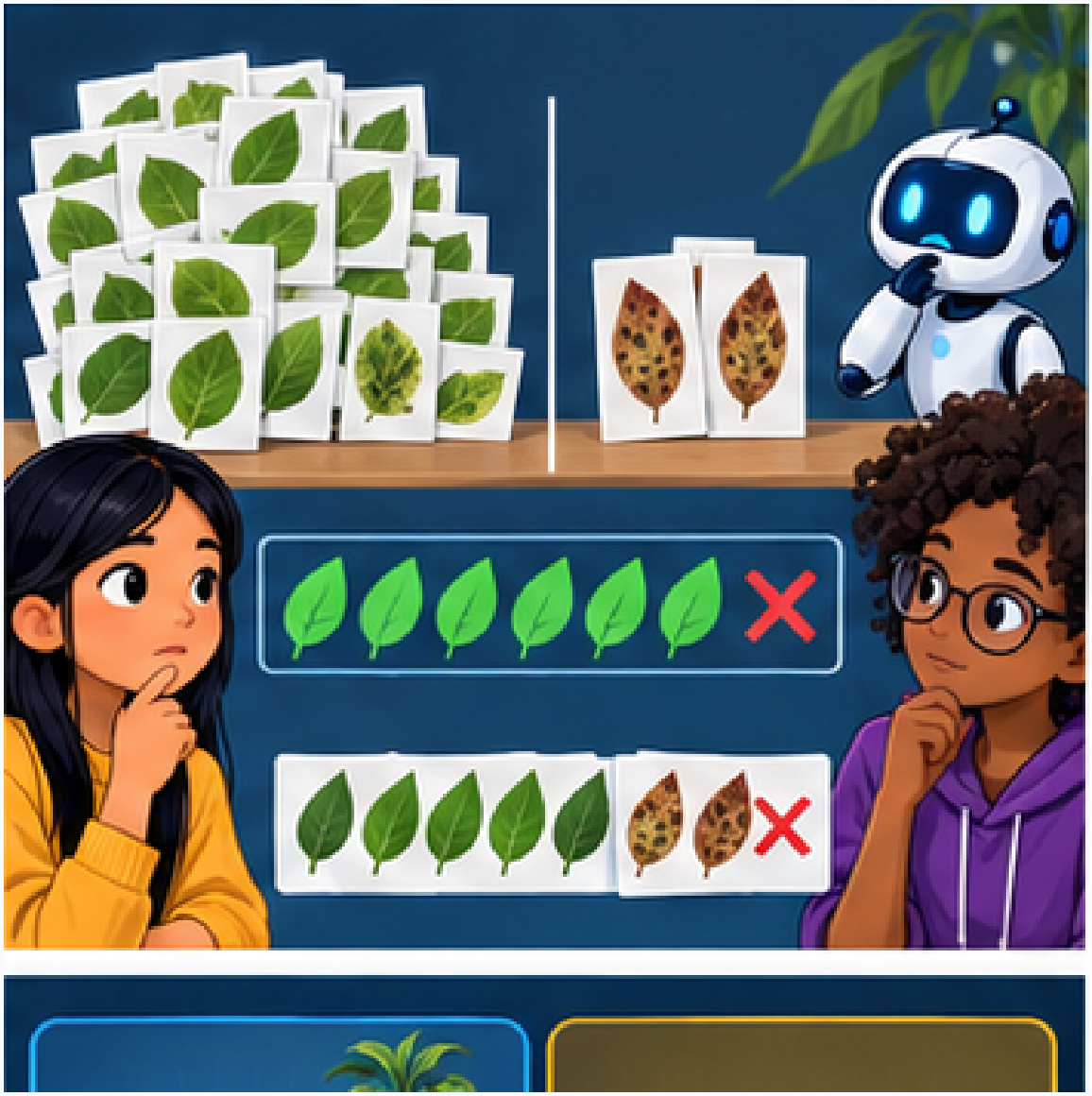
Notice the evidence

> Investigate the supplied Accuracy and Imbalance case. Separate observations, given facts and assumptions before choosing an explanation.

PICTURE ENTRY

Notice → Predict → Explain

Begin with visible **evidence**; do not explain more than the picture and information support.



Start with this puzzle

MYSTERY The team must use **accuracy** and imbalance to solve a model-building problem. What information do they have? What important **evidence** is still missing?

Name three visible details without interpreting them.

Predict the data, model, or evaluation decision being made.

Identify one hidden assumption that could change the result.

State what **evidence** would confirm or reject your **prediction**.

My **prediction** and supporting **evidence**:

A closer explanation

Accuracy counts all correct **prediction**s and divides by all evaluated examples. Each example has equal weight, so a common class contributes more to the score. A high result can hide failure on a rare class.

This week keeps the confusion matrix visible and checks both classes separately.

New fictional **training** data has 16 clear and 4 blocked labels. Its **majority baseline** predicts clear for every query; choose that rule from **training**, not evaluation labels.

A separate evaluation set also contains 16 clear and 4 blocked cases. A frozen candidate and the baseline are evaluated on the same 20 records.

Method and reference row	Predicted clear	Predicted blocked
Baseline reference clear	16	0
Baseline reference blocked	4	0
Candidate reference clear	12	4
Candidate reference blocked	0	4

Rows are references and columns are **prediction**s, ordered clear then blocked. Blocked is positive. Both methods get 16/20=80 percent **accuracy**. The baseline misses all four blocked cases.

The candidate finds all four but falsely flags four clear cases. The same score conceals a different tradeoff.

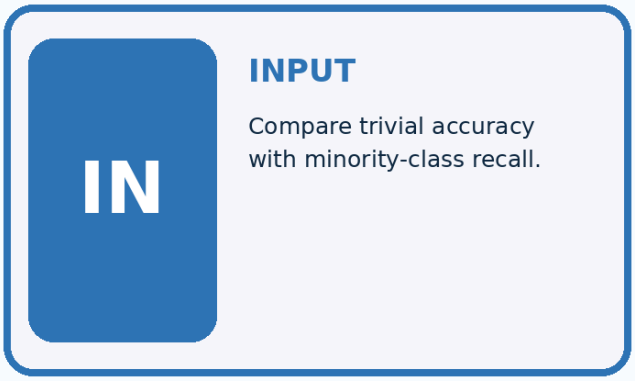
See how it works

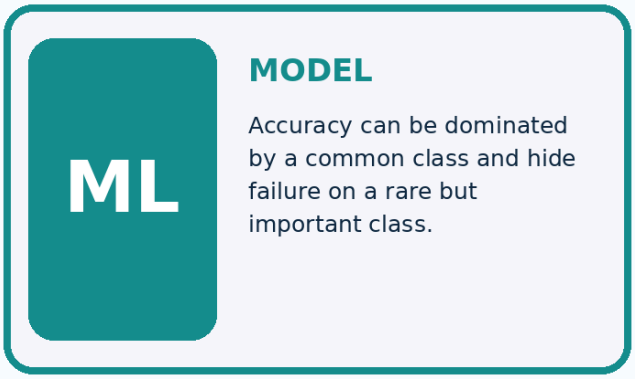
> Follow the mechanism for Accuracy and Imbalance. Connect each labelled input, deciding step and result.

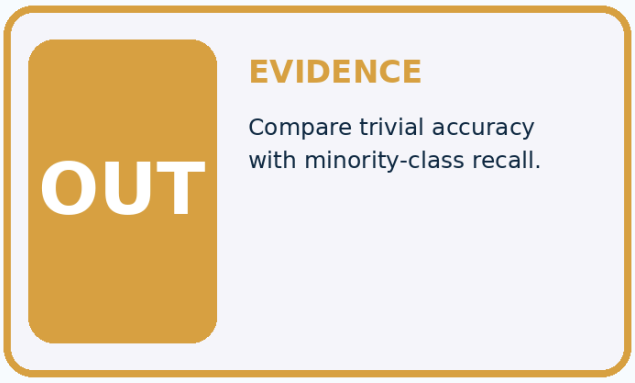
ML MECHANISM

Input → Model → Evidence

Trace the information, fitted behavior, and evaluation proof separately.







Question to explain

QUESTION Compare trivial **accuracy** with minority-class **recall**.

MODEL IDEA **Accuracy** can be dominated by a common class and hide failure on a rare but important class.

PROOF NEEDED Use an unseen case, visible calculation, error analysis, and a stated limitation.

How the mechanism works

UNDER THE HOOD Test set has 95 healthy and 5 diseased leaves. Always predicting healthy gives 95 ÷ 100 = 95% **accuracy**.

Draw a fresh data → model → **prediction** → evaluation trace:

A closer explanation

```
def scores(counts):
    clear_n = sum(counts[0])
    blocked_n = sum(counts[1])
    if clear_n == 0 or blocked_n == 0:
        raise ValueError("Need evaluation examples of both classes")
    correct = counts[0][0] + counts[1][1]
    accuracy = correct / (clear_n + blocked_n)
    clear_fraction = counts[0][0] / clear_n
```

```
    blocked_fraction = counts[1][1] / blocked_n
    balanced = (clear_fraction + blocked_fraction) / 2
    return accuracy, balanced

baseline = [[16, 0], [4, 0]]
candidate = [[12, 4], [0, 4]]
for name, counts in [("baseline", baseline), ("candidate", candidate)]:
    accuracy, balanced = scores(counts)
    print(name, round(100 * accuracy, 1), round(100 * balanced, 1))
```

The **output** is baseline 80.0 50.0, then candidate 80.0 87.5. In each **output** line the first percentage is ordinary **accuracy** and the second is **balanced accuracy**.

For the candidate, the clear-class correct fraction is 12/16=75 percent and the blocked-class fraction is 4/4=100 percent. Their equal average is 87.5 percent.

For the baseline, those fractions are 16/16=100 percent and 0/4=0 percent, averaging 50 percent. The code assumes a two-by-two table of nonnegative integer counts in the stated order.

It stops when either reference class is absent. Then report the observed counts and the unavailable class result; do not invent a denominator or silently substitute zero.

Connect the picture and the explanation

Point to the **input**, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

STUDENT LESSON · 4 OF 18

Compare the cases

> Compare the supplied cases for Accuracy and Imbalance. Identify the detail that changes the conclusion.

EVALUATION DESIGN

Four Tests, Four Kinds of Evidence

A system is not understood until it survives more than one comfortable example.


✓
NORMAL CASE
Expected data from the intended setting
◇
BOUNDARY CASE
Value near a threshold or class boundary
?
MISSING CASE
Unknown field, label, or necessary evidence
↗
SHIFTED CASE
New subgroup, device, condition, or time

NORMAL Expected data and expected behavior.

Keep this boundary clear

BOUNDARY Value close to a threshold, ambiguous label, or uncertain **prediction**.

MISSING Necessary field, label, source, or measurement is unavailable.

SHIFTED New lighting, species, device, subgroup, or time condition.

RELEASE RULE Passing one case never proves generalization. Predefine expected behavior for all four categories.

Which case most threatens today's claim, and why?

Change one thing in the pictured case

Change one thing

Change: The candidate’s three false alarms are corrected without changing its detections.

Predicted result

Prediction: Candidate **accuracy** becomes20/20 with spotted **recall** still100% in this toy set.

Why the result follows

Reason: False alarms move from FP to TN; neither TP nor FN changes.

STUDENT LESSON · 5 OF 18

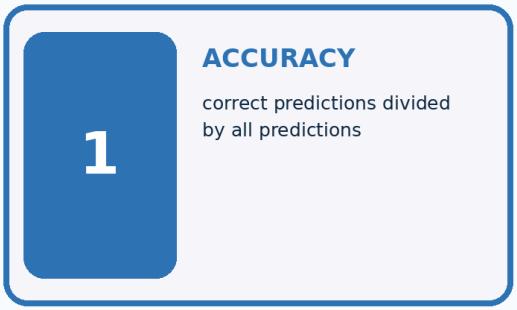
Words that unlock the idea

> Use the vocabulary of Accuracy and Imbalance to explain a concrete decision, rather than only memorising definitions.

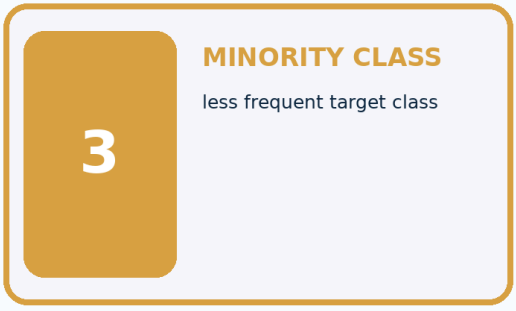
VOCABULARY

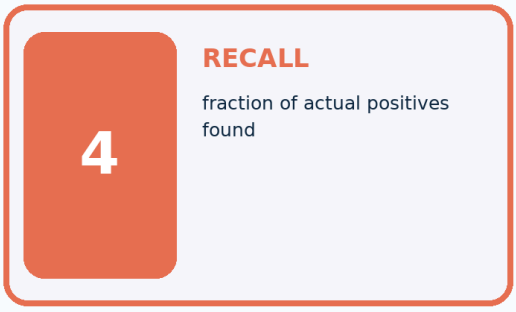
Words You Can Calculate and Defend

A definition matters when it lets you interpret a new dataset, model, or result.









Term	Precise meaning	My ML example
accuracy	correct predictions divided by all predictions	
class imbalance	classes occur in very different amounts	
minority class	less frequent target class	
recall	fraction of actual positives found	

MISCONCEPTION TO REPAIR A classifier with 95% **accuracy** is excellent in every practical situation.

Repair the claim using a definition, calculation, or test record:

Words used in the closer explanation

Word	Meaning
Class imbalance	Unequal numbers of examples in different classes
Majority baseline	A rule that always predicts the most common training label
Class-specific correct fraction	Correct predictions for one reference class divided by the number of examples of that class
Balanced accuracy	The average of the class-specific correct fractions with equal weight for each class

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STUDENT LESSON · 6 OF 18

Follow a worked example

> Follow the worked trace for Accuracy and Imbalance, predict a step, then compare with its explanation.

WORKED EVIDENCE

Reason Step by Step

Predict first. Then check each step. A correct answer without visible **evidence** cannot be audited.

WEEK 16 • WORKED EVIDENCE

1. Test set has 95 healthy and 5 diseased leaves.
2. Always predicting healthy gives 95 ÷ 100 = 95% accuracy.
3. It finds 0 of 5 diseased leaves, so diseased recall = 0%.
4. Report class counts and recall; redesign data or use only where misses are acceptable.

Step 1

STEP 1 Test set has 95 healthy and 5 diseased leaves.

Step 2

STEP 2 Always predicting healthy gives 95 ÷ 100 = 95% **accuracy**.

Step 3

STEP 3 It finds 0 of 5 diseased leaves, so diseased **recall** = 0%.

Step 4

STEP 4 Report class counts and **recall**; redesign data or use only where misses are acceptable.

CHECK Verify the denominator, unit, split, class definition, and data version before interpreting a score.

Recalculate, retrace, or restate the **evidence** independently:

A closer explanation

Same overall score different class results

SUMMARY	ALWAYS CLEAR BASELINE	FROZEN CANDIDATE
Overall correct	16 of 20 is 80 percent	16 of 20 is 80 percent
Reference clear correct	16 of 16 is 100 percent	12 of 16 is 75 percent
Reference blocked correct	0 of 4 is 0 percent	4 of 4 is 100 percent

Overall accuracy weights classes by their number of evaluation examples. Balanced accuracy gives the two class-specific fractions equal weight.

Both methods score 80 percent. Explain which cases the baseline fails, how the candidate changes the errors, and why their balanced accuracies differ.

Worked explanation · Accuracy and Imbalance

Balanced accuracy is a metric choice, not a repair to the classifier or a guarantee of safety. It gives classes equal influence in this summary; it does not encode the actual consequences of false alarms and missed cases.

Four blocked evaluation examples are also too few for a strong general claim.

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: Accuracy and Imbalance

Show how a common healthy class can hide complete failure to detect spotted leaves.

The facts we will use

Supplied facts

Toy evaluation has 18 healthy and 2 spotted leaves. Always-healthy predicts all healthy. A candidate hasTP2, FN 0, FP 3, TN 15.

Supplied facts

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

AI ACADEMY / AI-4/16

Candidate decisions

	predicted spotted	predicted healthy
actual spotted	2	0
actual healthy	3	15

Read a cell using both its row and column labels.

Follow the labelled picture with the matching explanation. Open the vector diagram

AI ACADEMY / AI-4/16

Accuracy and Imbalance

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Score trivial baseline

Always healthy is 18/20=90% accurate but detects0/2 spotted leaves.



2. Score candidate

Candidate accuracy is 17/20=85%; spotted recall is 2/2=100%.



3. Compare consequences

Candidate catches both spotted leaves while producing three false alarms; choose with stated costs and review needs.

CHANGE ONE THING

The candidate's three false alarms are corrected without changing its detections.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Score trivial baseline	Always healthy is 18/20=90% accurate but detects0/2 spotted leaves.
2. Score candidate	Candidate accuracy is 17/20=85%; spotted recall is 2/2=100%.
3. Compare consequences	Candidate catches both spotted leaves while producing three false alarms; choose with stated costs and review needs.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

Why the result follows

Read the picture in order. Candidate catches both spotted leaves while producing three false alarms; choose with stated costs and review needs. The arrows show which recorded **input** or intermediate result each decision uses.

Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

What this example does not establish

Keep the stated limit

Perfect performance on twenty supplied cases does not demonstrate dependable detection in other plants or settings.

Keep the stated limit

Read the labelled picture: Why can the 90% baseline still be unsuitable for a detection task? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Keep the stated limit

Change one thing: The candidate's three false alarms are corrected without changing its detections. Predict the result before checking the explanation. What remains the same?

STUDENT LESSON · 7 OF 18

Find and repair a mistake

> Find the unsupported step in the Accuracy and Imbalance claim and explain a repair using the supplied evidence.

Find the first error

ERROR DETECTIVE

Compare → Diagnose → Repair

Do not patch the answer. Find the earliest unsupported assumption, data problem, or experiment error.

!

COMPARE • DIAGNOSE • REPAIR

Tempting claim: A classifier with 95% accuracy is excellent in every practical situation. Inspect the dataset, split, calculation, error type, subgroup, or experiment record; identify one testable cause; change one factor; rerun the same evidence.

Evidence record	Expected	Actual	Likely cause / next test
normal case	passes		
boundary case	defined behavior		
missing case	safe fallback		
shifted case	measured limit		

CLAIM TO CHALLENGE A classifier with 95% **accuracy** is excellent in every practical situation.

Diagnose one cause and name the cleanest next experiment:

A closer explanation

Check the claim against the facts

Compare methods on the same appropriate evaluation examples and report the **training**-derived baseline. Include ordinary **accuracy**, each class denominator, false positives and false negatives.

An equal-class summary adds information but cannot choose the acceptable tradeoff for the task. Keep development decisions separate from a final test.

Check the claim against the facts

A result of 4/4 correct blocked cases is 100 percent on those four examples, not proof of perfect future detection. If the evaluated set contains no blocked cases, it cannot measure blocked-case performance at all.

Next week separates precision and **recall** by asking which denominator answers each question.

A team advertises 80 percent accuracy but its classifier predicts clear on every record. What additional evidence would you request, and what can you already conclude about the four blocked cases in our worked evaluation?

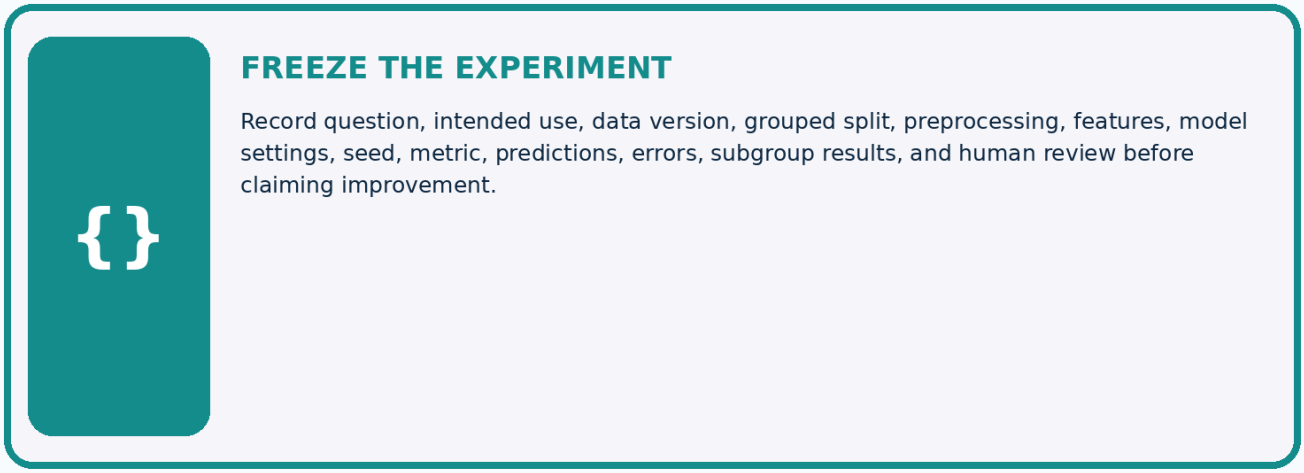
Practise together

> Practise Accuracy and Imbalance with support, then explain the deciding step in your own words.

ML STUDIO

Specify Before You Run

A trustworthy result can be reproduced from the recorded data, code, settings, and evaluation rule.



Experiment record

Question to explain

QUESTION + intended use + success threshold

DATA version + grouped split + leakage check

CODE/pre**process**ing + features + model + settings + seed

METRICS + positive class + units + denominators

PREDICTIONS + errors + subgroup slices + limitations

HUMAN reviewer + release, revise, limit, or stop decision

Write the minimum record needed to reproduce today's result:

A closer explanation

Consider a separate illustrative set with 8 clear and 8 blocked cases. Suppose the same within-class fractions hold: the baseline matrix is [[8,0],[8,0]] and the candidate matrix is [[6,2],[0,8]].

Ordinary **accuracy** is now 50 percent versus 87.5 percent. **Balanced accuracy** remains 50 percent versus 87.5 percent. The class mix changed the overall score even though the within-class fractions stayed fixed.

This is an arithmetic comparison, not proof that a real classifier keeps the same behavior after a population change. Report sample size, class counts, full matrix and how the evaluation data was chosen.

Do not remove inconvenient cases or duplicate rare evaluation cases just to improve a headline number. If **training** is resampled, keep that separate from evaluation and document it.

Practise with less help: Compare both class fractions

Use workbook W2 for the connected example "**Accuracy** and **class imbalance**". First fill the **input** and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan	Do	Check
What is given? Which rule or procedure applies?	Record each intermediate step.	Does the result follow? What remains unknown?

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

Check your reasoning

Use the worked case for **Accuracy** and **class imbalance**. Proposed explanation: "High **accuracy** means both classes work well". Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Answer on your own before discussion. If you are unsure, say which step you need help with.

Explain the picture

Why can the 90% baseline still be unsuitable for a detection task? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

> **My explanation using a pictured step and the given facts**

This is supported practice. Keep the separately named independent case for an attempt before feedback.

STUDENT LESSON · 9 OF 18

Run the investigation

> Investigate Accuracy and Imbalance. Record the prediction, actual observation or labelled paper trace, explanation and next test separately.

EVIDENCE LAB

Predict → Run → Calculate → Interpret

Keep the test conditions fixed, record every result, and change only one planned factor.

LAB

PREDICT → RUN → CALCULATE → INTERPRET

Imbalance Simulation: compare majority prediction with a model using accuracy, recall, and the actual error costs.

1. Write the hypothesis and expected **evidence** before running.

2. Imbalance Simulation: compare majority **prediction** with a model using **accuracy**, **recall**, and the actual error costs.

3. Run one normal, one boundary, one missing, and one shifted case.

4. Calculate the metric from raw counts or row-level errors.

5. Interpret what changed, what did not, and what remains unsafe or unknown.

Case · Prediction · Actual · Calculation / evidence · Meaning / next test

> **Case:** Normal

Prediction: _____

Actual: _____

Calculation / evidence: _____

Meaning / next test: _____

Case: Boundary

Prediction: _____

Actual: _____

Calculation / evidence: _____

Meaning / next test: _____

Case: Missing

Prediction: _____

Actual: _____

Calculation / evidence: _____

Meaning / next test: _____

Case: Shifted

Prediction: _____

Actual: _____

Calculation / evidence: _____

Meaning / next test: _____

Plan → try → check

Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

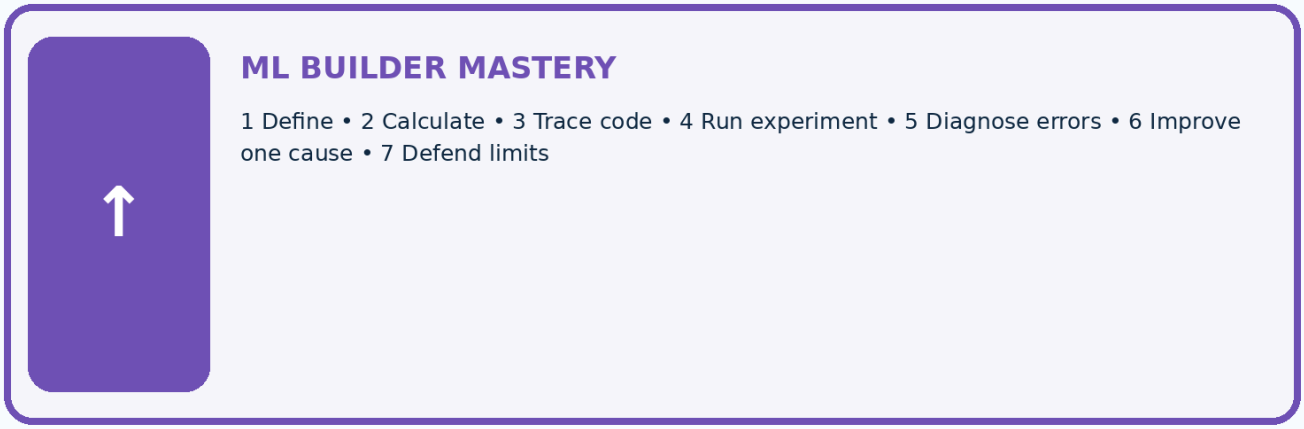
Build your understanding

> Move from recall to tracing, diagnosis and transfer in Accuracy and Imbalance. Explain why each step is justified.

MASTERY

From Knowing the Word to Owning the Idea

Move upward only when you can produce **evidence** without copying the example.



DEFINE Use all four terms precisely: **accuracy**, **class imbalance**, **minority class**, **recall**.

CALCULATE Reproduce the worked **evidence** with units and denominator.

TRACE Explain the data, code or decision path.

DIAGNOSE A classifier with 95% **accuracy** is excellent in every practical situation.

TRANSFER Solve a fresh dataset or model case.

IMPROVE Change one cause and rerun fixed **evidence**.

DEFEND State intended use, metric, limitation, and human responsibility.

MASTERY RULE A coached calculation becomes mastery only after a different case is completed independently.

STUDENT LESSON · 11 OF 18

Connect your project

> Apply this week's idea to Responsible Plant Classifier. Keep an annotated decision and the evidence supporting it.

CUMULATIVE PROJECT

Responsible Plant Classifier

Every week adds one reviewable artifact to the same **evidence**-based capstone.

RESPONSIBLE PLANT CLASSIFIER

Check Plant Classifier class balance and select metrics that keep the spotted-leaf class visible.

THIS WEEK'S CHECKPOINT Check Plant Classifier class balance and select metrics that keep the spotted-leaf class visible.

Which requirement or risk does this artifact address?

Which dataset, split, code, and metric version does it use?

What normal and shifted cases were tested?

What remains uncertain or weak?

What decision still belongs to a teacher or qualified human?

RESPONSIBLE RELEASE Never use this classroom prototype for medical, pesticide, agricultural, or other consequential decisions.

My **evidence**-linked project decision:

Evidence for your project

For your Responsible Plant Classifier, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

STUDENT LESSON · 12 OF 18

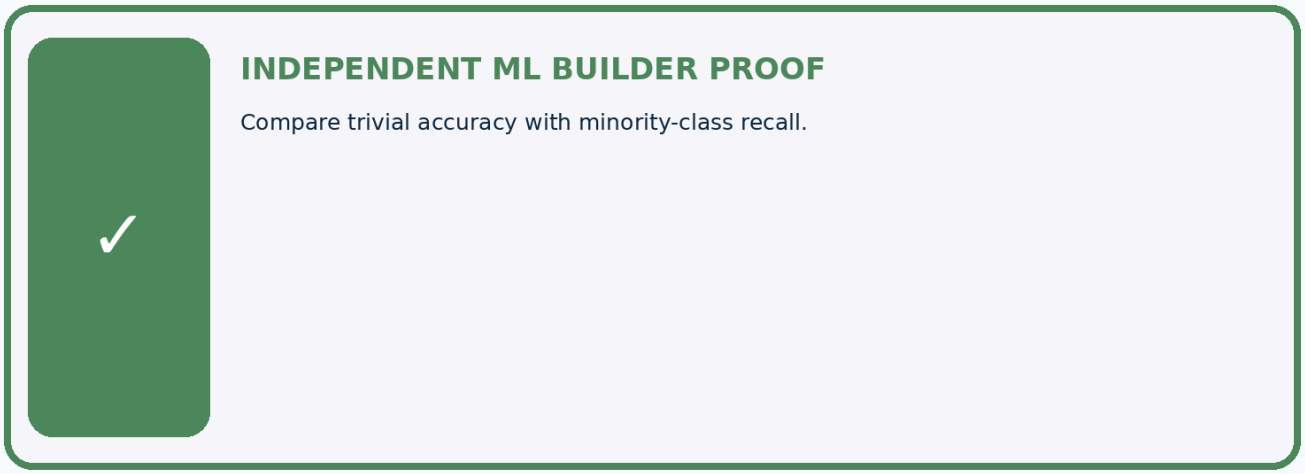
Show what you know

> Use a genuinely new case for independent evidence. Previously discussed examples remain practice; use the separately named independent case before feedback.

INDEPENDENT PROOF

Explain • Calculate • Transfer • Defend

Complete this with a fresh case after guided practice and compare it with the mastery criteria.



What success looks like

REQUIRED PROOF Compare trivial **accuracy** with minority-class **recall**.

- ☐ Define the task, feature/target roles, and intended-use boundary.
- ☐ Show the calculation, code trace, or experiment record.
- ☐ Test a normal case and one revealing boundary, missing, or shifted case.
- ☐ Diagnose one failure without changing several factors at once.
- ☐ Name a limitation, fairness concern, and the human-review action.

Independent **evidence** and defense:

My level: ☐ Not yet ☐ With one hint ☐ Independently ☐ I can teach and defend it

Connect to the next lesson

NEXT BRIDGE Carry the experiment **evidence** into: Precision and **Recall**.

STUDENT LESSON · 13 OF 18

Study a complete case

> Revisit the complete Accuracy and Imbalance case for supported practice. Repeated examples do not become fresh assessments.

Week 16 Worked case

Use the supplied details to practise the weekly concept before attempting the independent question. This case extends the illustrated lesson.

The situation

A dataset has 95 healthy and 5 spotted. Model predicts healthy for all.

The question

Calculate and critique **accuracy**.

Worked explanation

95/100=95%, but spotted **recall**=0/5=0%. Majority performance hides the detection failure.

Explain it in your own words

> Which detail supports the answer, and why does it matter?

Write or draw your reasoning and evidence.

Trace the supplied case visually

The majority can hide a detection failure

Evidence or stage	Value or result
Healthy, predicted healthy	95
Spotted, predicted healthy	5

A dataset has 95 healthy and 5 spotted. Model predicts healthy for all.

Calculate and critique **accuracy**.

95/100=95%, but spotted **recall**=0/5=0%. Majority performance hides the detection failure.

STUDENT LESSON · 14 OF 18

Try a new case independently

> Attempt the reserved new case before feedback. Record support; the earlier examples below remain additional practice.

Independent case: Changed independent case

Validation has90 healthy and10 spotted cases. Always-healthy predicts all healthy. Another model has TP7,FN3,FP8,TN82. Compare **accuracy** and spotted **recall**.

> **My independent explanation, trace and limits**

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Week 16 Independent application

Close the worked explanation. Apply the idea to this different question without copying.

A classifier predicts healthy for all 100 plants; 95 are healthy and 5 diseased. Compute **accuracy** and diseased **recall**.

My answer, with a trace, calculation or supporting **evidence**

> **Write or draw your reasoning and evidence.**

Create and test

Change one relevant detail. Predict the result before testing. Include a boundary or missing-information case when it helps reveal a limit.

My changed **input** and expected result

> **Write or draw your reasoning and evidence.**

Connect to your project

Check Plant Classifier class balance and select metrics that keep the spotted-leaf class visible.

The **evidence** I will keep

Write or draw your reasoning and evidence.

Your teacher has the independent answer and scoring criteria. Complete the matching workbook week to keep your practice **evidence** together.

Transfer practice from the original programme

Model A is always healthy on 96 healthy and 4 spotted leaves. Model B finds 3 spotted but falsely flags 5 healthy. Compare **accuracy** and spotted detection.

Explain the mechanism, evidence and one limit.

Use feedback to improve

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed.

A later check uses a changed case, not a copied answer.

STUDENT LESSON · 15 OF 18

Your lesson route

Use this map to navigate Accuracy and Imbalance, connect the goals to tasks and keep one evidence record for each attempt.

Start with this question: Show how a common healthy class can hide complete failure to detect spotted leaves.

By the end, I can	How I will show it
Explain why overall accuracy can hide failure on a less common class	Explain the deciding step, show a trace or test, and state one limit.
Compare a training-majority baseline with class-specific results	Explain the deciding step, show a trace or test, and state one limit.
Calculate an equal-class average and report its limits alongside the confusion matrix	Explain the deciding step, show a trace or test, and state one limit.

Remember → watch and explain → try with help → investigate → try independently → keep project **evidence**. Your teacher may spread this lesson across several classes.

Keep your first **prediction**. Compare it with what the **evidence** shows and explain any change.

STUDENT LESSON · 16 OF 18

Finish and return

> Explain Accuracy and Imbalance without notes, identify a remaining question and plan a later check with a changed case.

Close your notes. Explain this goal in your own words: Explain why overall **accuracy** can hide failure on a less common class

My deciding step, evidence and one remaining question

If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an **input** or assumption and justify your **prediction**.

Save your workbook **evidence**. At the next review, try a different case before opening your old answer.

STUDENT LESSON · 17 OF 18

Supported practice, repair and portfolio

> Attempt supported questions before using hints or feedback. Review a missing building block, then keep the completed work.

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Show how a common healthy class can hide complete failure to detect spotted leaves.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? Candidate catches both spotted leaves while producing three false alarms; choose with stated costs and review needs.

1. Compare consequences

2. Score candidate

3. Score trivial baseline

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Compare consequences Candidate catches both spotted leaves while producing three false alarms; choose with stated costs and review needs. Read the picture in order.

Candidate catches both spotted leaves while producing three false alarms; choose with stated costs and review needs. The arrows show which recorded **input** or intermediate result each decision uses.

Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Practice 2: explain the deciding evidence

Changed case · predict first

Changed-case practice: The candidate’s three false alarms are corrected without changing its detections. Which conclusion is supported by the supplied facts?

1. Candidate **accuracy** becomes20/20 with spotted **recall** still100% in this toy set.

2. This one example guarantees correct results for every future **input**.

3. We can decide the result without examining the changed **input** or the procedure.

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Candidate **accuracy** becomes20/20 with spotted **recall** still100% in this toy set. False alarms move from FP to TN; neither TP nor FN changes.

Repair a missing building block

Review Confusion Matrix (ai-4/15). Goal: Build a labelled confusion matrix from matched reference and **prediction** pairs. Short practice: Why is adding TP alone without reducing FN an incorrect revision?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Train a Classifier (ai-4/13). Goal: Connect **training**-only scaling and neighbour **prediction** in one pipeline. Short practice: Which pipeline record would help another team reproduce the result?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Baselines First (ai-4/10). Goal: Fit a majority label baseline using **training** labels only. Short practice: Can the candidate's **training accuracy** replace this validation comparison?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Predict, run and debug

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual **output** or error, change one **input**, then run again.

Keep code, **prediction**s, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

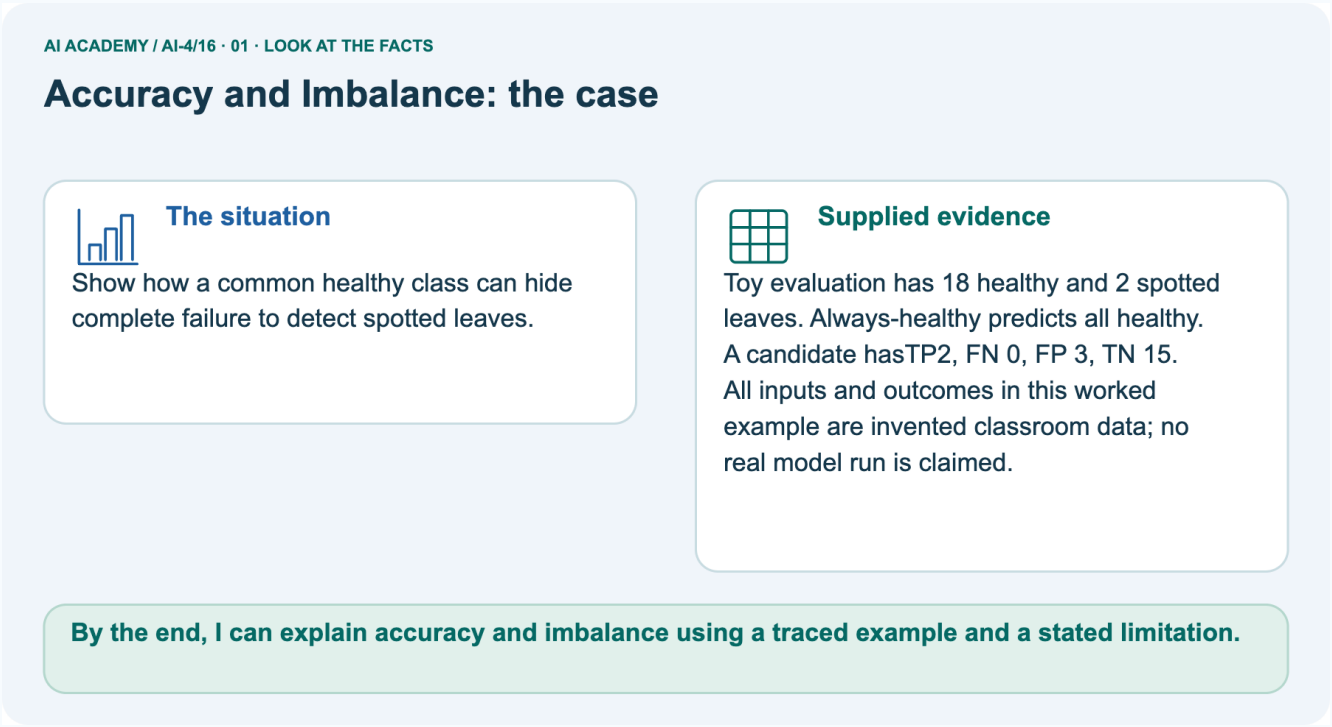
Drafts stay in this browser when storage is available; clear drafts on a shared device.

STUDENT LESSON · 18 OF 18

Visual concept journey

> Explain each Accuracy and Imbalance graphic using its labels, deciding evidence and relevant limitation.

01. Accuracy and Imbalance: the case



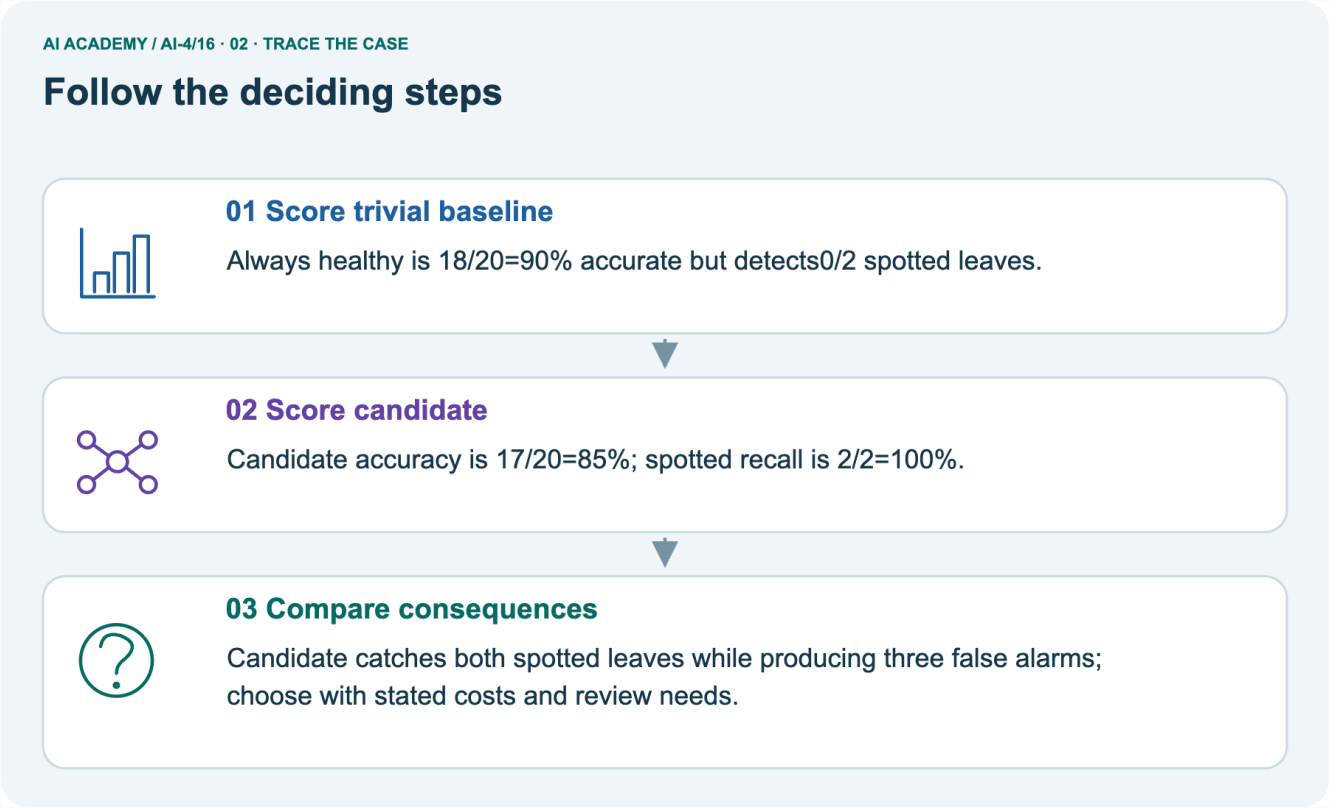
Follow the labelled picture with the matching explanation. Open the vector diagram

Show how a common healthy class can hide complete failure to detect spotted leaves. Supplied facts: Toy evaluation has 18 healthy and 2 spotted leaves. Always-healthy predicts all healthy.

A candidate hasTP2, FN 0, FP 3, TN 15. All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

> Point to the given fact that matters most. Explain why it matters.

02. Follow the deciding steps

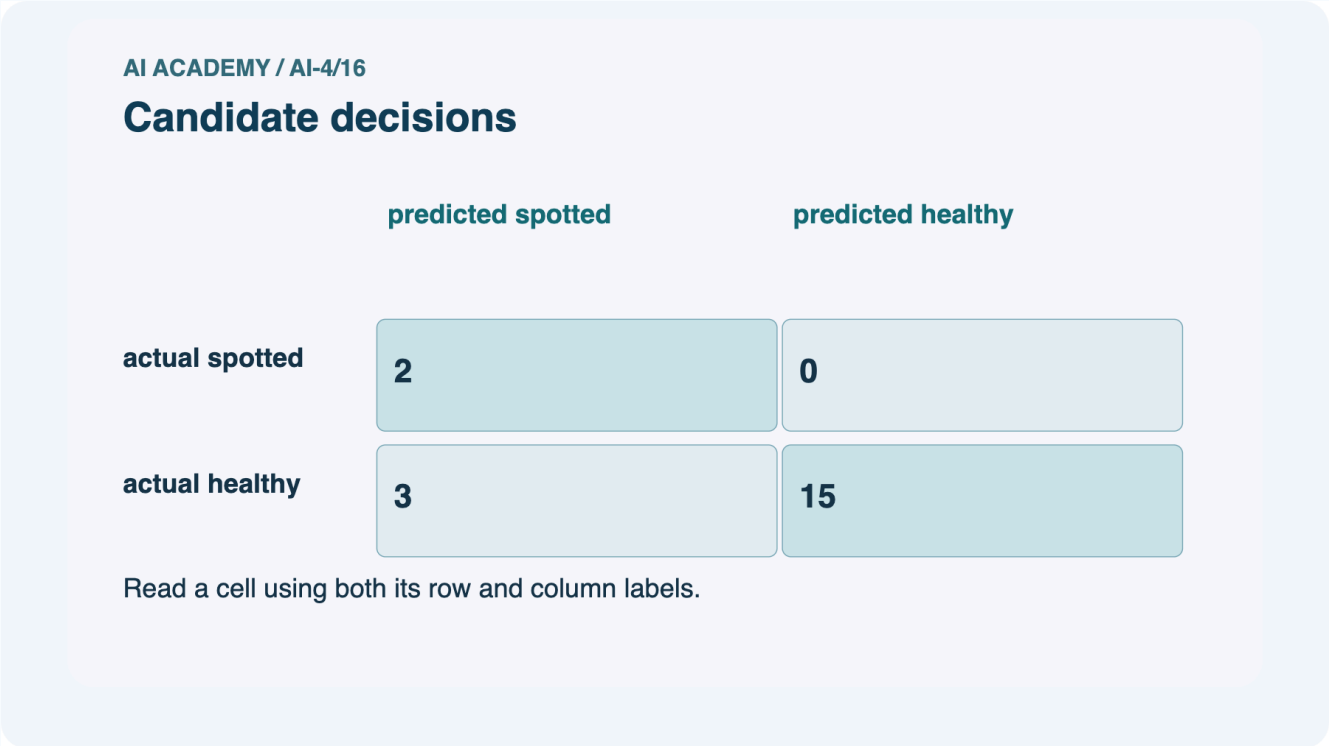


Follow the labelled picture with the matching explanation. Open the vector diagram

Score trivial baseline: Always healthy is 18/20=90% accurate but detects0/2 spotted leaves. → Score candidate: Candidate **accuracy** is 17/20=85%; spotted **recall** is 2/2=100%. → Compare consequences: Candidate catches both spotted leaves while producing three false alarms; choose with stated costs and review needs.

> Point to each numbered stage and explain which recorded **input** it uses.

03. Candidate decisions

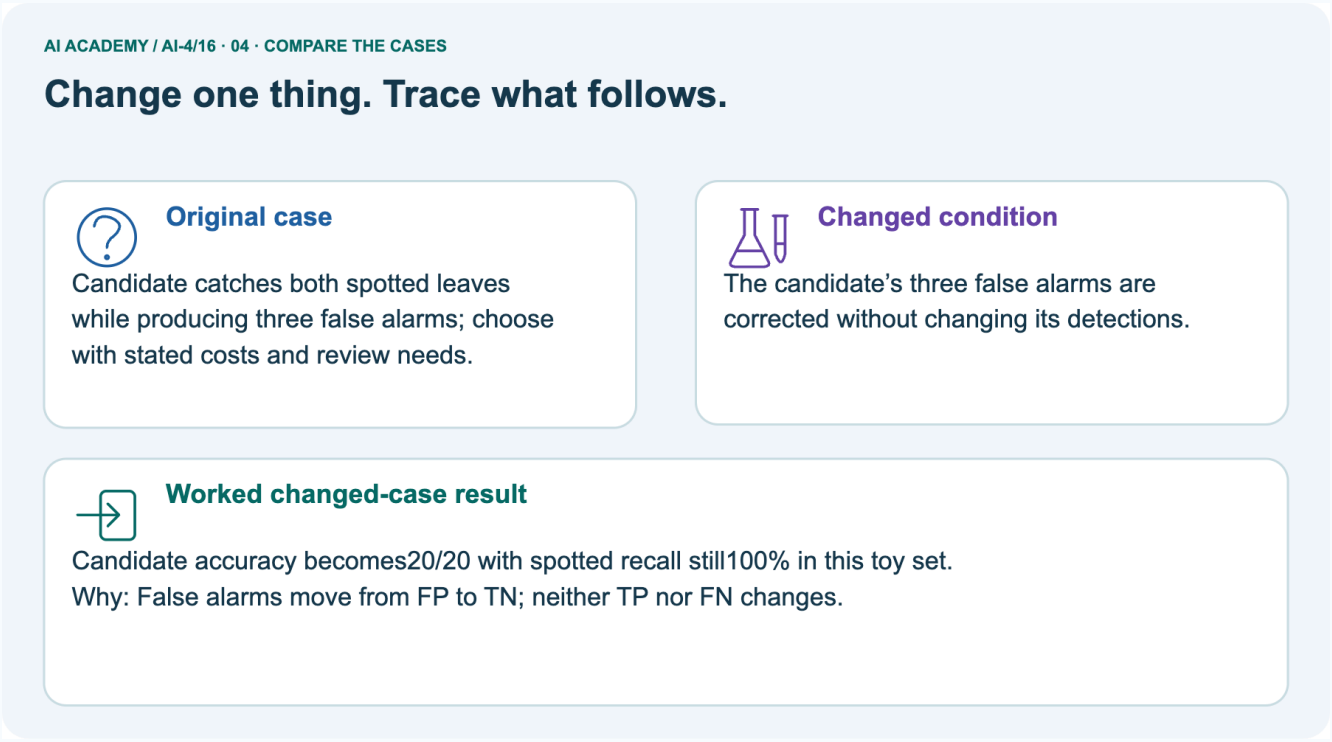


Follow the labelled picture with the matching explanation. Open the vector diagram

Candidate decisions. actual spotted: predicted spotted = 2, predicted healthy = 0. actual healthy: predicted spotted = 3, predicted healthy = 15.

> Read the labels and values. Explain the relationship the model shows.

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

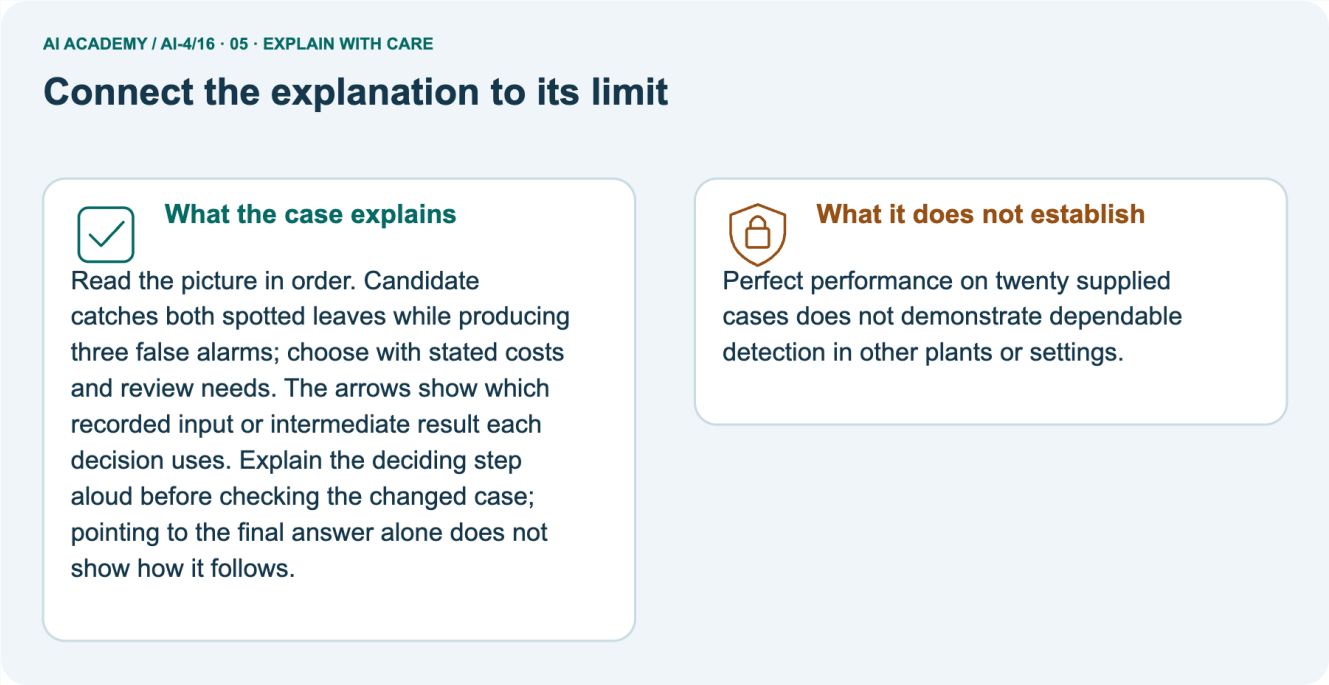
Original case: Candidate catches both spotted leaves while producing three false alarms; choose with stated costs and review needs.

Changed condition: The candidate's three false alarms are corrected without changing its detections. Expected result: Candidate **accuracy** becomes20/20 with spotted **recall** still100% in this toy set.

Reason: False alarms move from FP to TN; neither TP nor FN changes.

> Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

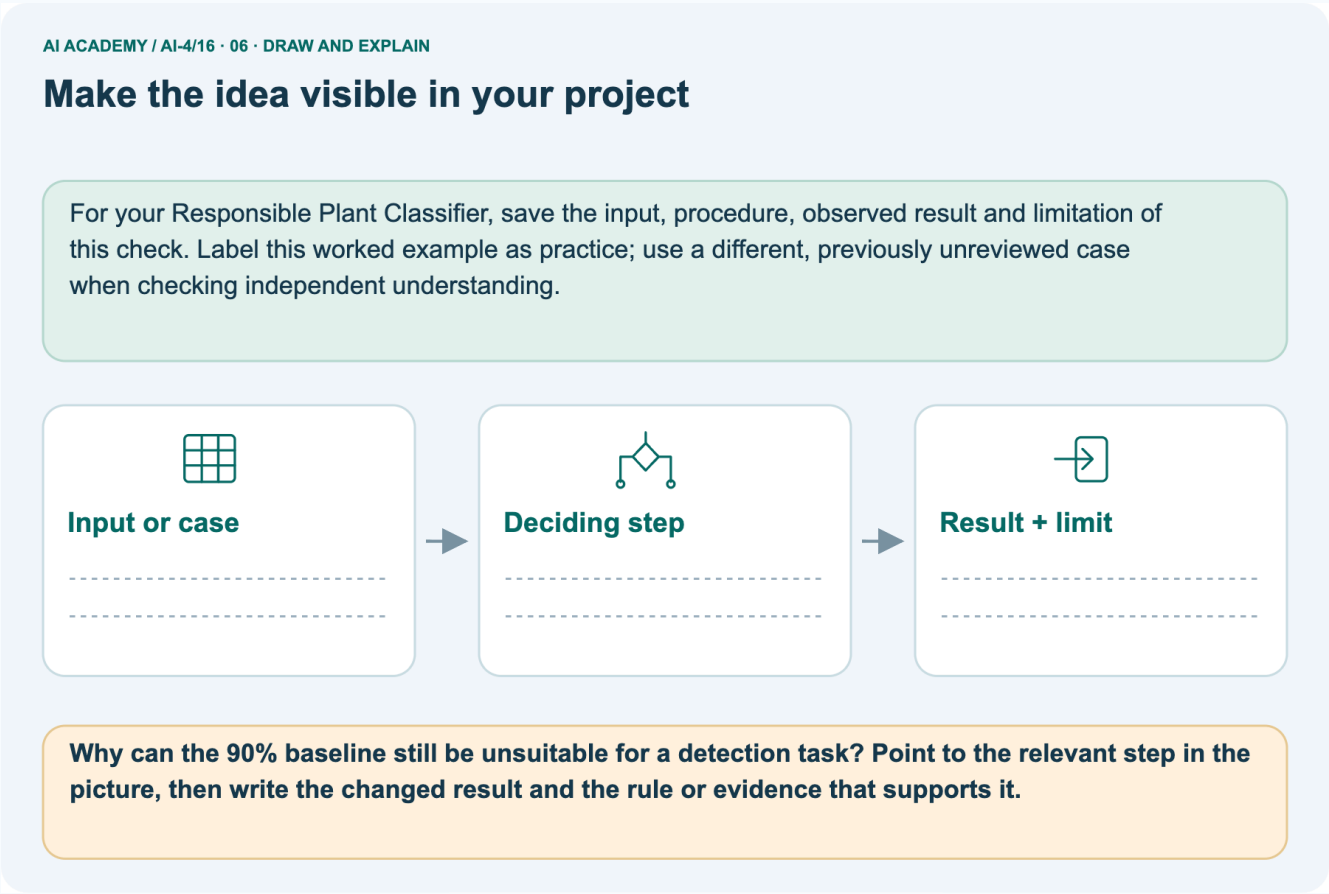
Read the picture in order. Candidate catches both spotted leaves while producing three false alarms; choose with stated costs and review needs. The arrows show which recorded **input** or intermediate result each decision uses.

Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Limit: Perfect performance on twenty supplied cases does not demonstrate dependable detection in other plants or settings.

> Explain the case, then point to a limit on the claim.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For your Responsible Plant Classifier, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding. Draw your **input**, deciding step, result and limit.

Supported question: Why can the 90% baseline still be unsuitable for a detection task? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

> Draw your project trace and justify the deciding step using **evidence**.

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.